

PROJECT1: 2-STAGE OPAMP DESIGN

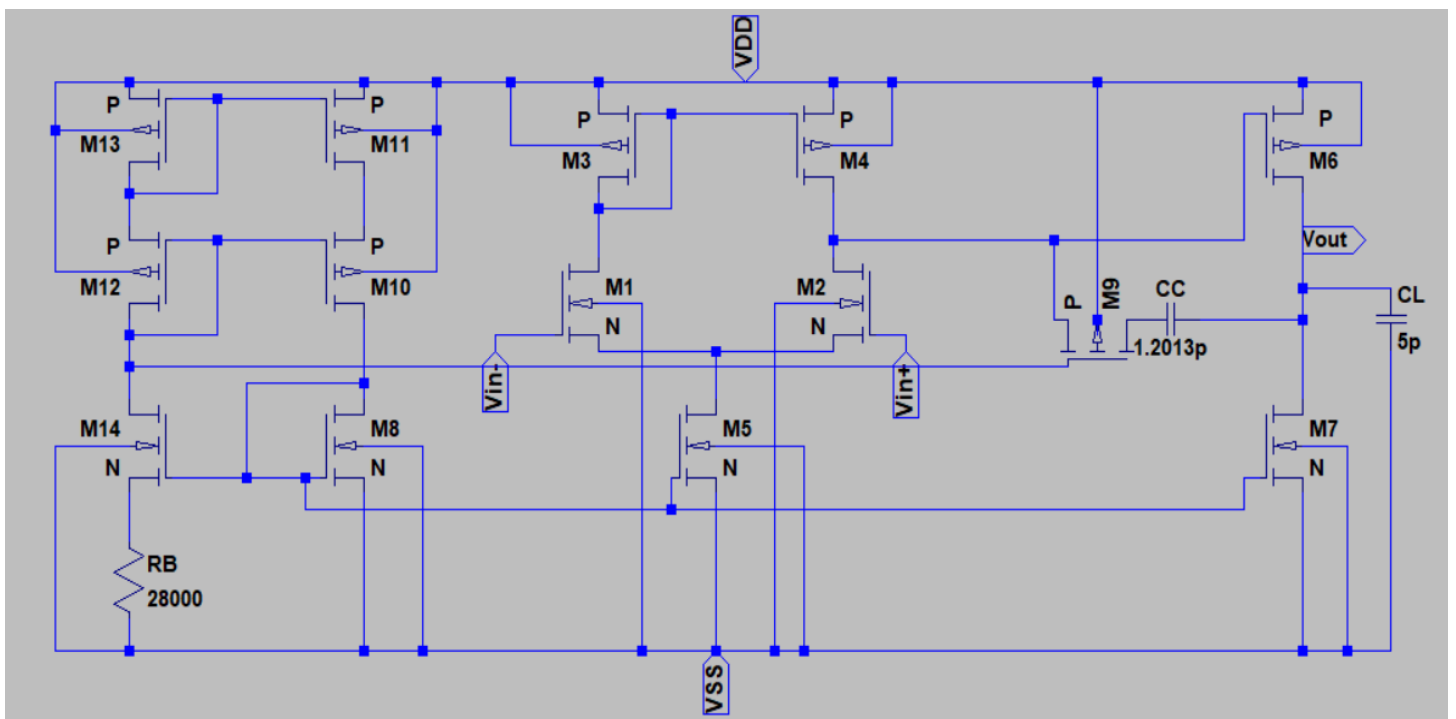
#specs

VDD = 2.5 # V
 VSS = -2.5 # V
 SR = 5.1e6 # +5.1/-5.1 V/us (original spec: 5 V/us)
 CL = 5e-12 # 5 pF
 Vin_cm_max = 2.1 # V
 Vin_cm_min = -1.3 # V
 Vout_max = 2.2 # V
 Vout_min = -2.2 # V
 GBW = 5.5e6 # 5,5 MHz (original spec: 5 MHz)
 PM = 60
 Gain_dB = 80 # dB
 Sn_sqrt = 30e-9

Cc = 1.2112247672579234e-12 F
 RB = 27716.230218857112 ohm
 S1 (W/L)1 = 1.5584
 S2 (W/L)2 = 1.5584
 S3 (W/L)3 = 1.6539
 S4 (W/L)4 = 1.6539
 S5 (W/L)5 = 0.5789
 S6 (W/L)6 = 16.9624
 S7 (W/L)7 = 2.9688
 S8 (W/L)8 = 0.5789
 S9 (W/L)9 = 3.3078
 S10 (W/L)10 = 3.3078
 S11 (W/L)11 = 3.3078
 S12 (W/L)12 = 3.3078
 S13 (W/L)13 = 3.3078
 L6 = 4.1110e-06 m
 W6 = 6.9733e-05 m

Sizes	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14
W/L	1.55	1.55	1.65	1.65	0.57	16.96	2.85	0.55	3.22	3.22	3.22	3.22	3.22	2.21
W (um)	1.55	1.55	1.65	1.65	0.57	69.73	2.85	0.55	3.22	3.22	3.22	3.22	3.22	2.21
L (um)	1	1	1	1	1	4.11	1	1	1	1	1	1	1	1

Circuit Schematic

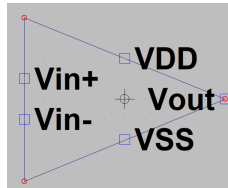


DESIGN PROCEDURE:

1) Modified Specifications: To account for non-idealities, I slightly increased the specs during design: **GBW** from 5 MHz to **5.5 MHz** and **SR** from 5 V/ μ s to **5.1 V/ μ s**, ensuring reliable performance in simulation.

2) Analytical Calculations: I used a Python notebook to perform transistor sizing and bias calculations, following the analytical formulas discussed in class. Key design parameters such as slew rate, GBW, input-referred noise, and phase margin were used to derive the compensation capacitor C_c , branch currents, and transistor dimensions (e.g., M6) to meet frequency and stability targets. The input pair and current mirrors were sized based on overdrive voltages and current scaling, and the bias resistor R_B was calculated to ensure proper gate biasing. All steps and values are documented in the attached notebook.

3) After sizing the transistors and calculating R_{BR_BRB} and C_{cC_cCc} , I sketched the schematic and created a 5-pin op-amp symbol (V_{in+} , V_{in-} , VDD, VSS, V_{out}) for use in all simulations.



4) simulations:

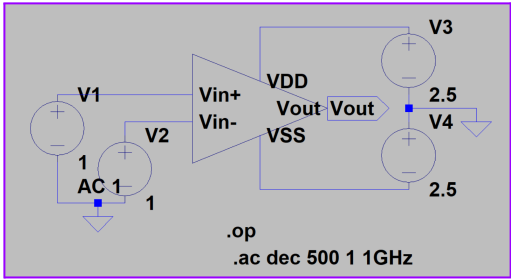
4-1- Operating Point Analysis (.op)

The **.op** simulation confirmed correct biasing: all transistors were in saturation ($C_{gd}=0$), except **M9**, which was intentionally in the linear region for zero-nulling. This validated the operating regions before further tests.

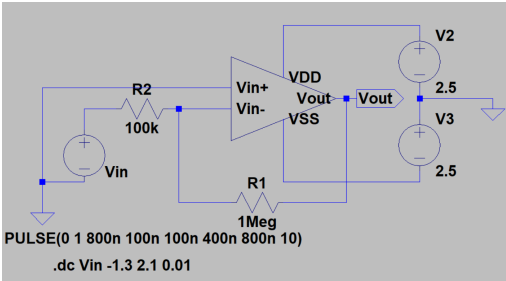
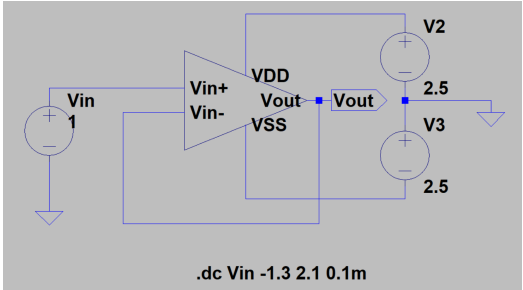
--- MOSFET Transistors ---						Name:	m:xl:6	m:xl:4	m:xl:3	m:xl:14	m:xl:8
Name:	m:xl:13	m:xl:12	m:xl:11	m:xl:10	m:xl:9	Model:	xl:p	xl:p	xl:p	xl:n	xl:n
Id:	-6.22e-06	-6.22e-06	-6.22e-06	-6.22e-06	-7.93e-21	Id:	-3.26e-05	-3.16e-06	-3.16e-06	6.22e-06	6.22e-06
Vgs:	-1.20e+00	-1.21e+00	-1.20e+00	-1.20e+00	-1.20e+00	Vgs:	-1.20e+00	-1.20e+00	-1.20e+00	8.81e-01	1.05e+00
Vds:	-1.20e+00	-1.21e+00	-1.20e+00	-2.75e+00	2.98e-08	Vds:	-1.79e+00	-1.20e+00	-1.20e+00	2.42e+00	1.05e+00
Vbs:	0.00e+00	1.20e+00	0.00e+00	1.20e+00	1.20e+00	Vbs:	0.00e+00	0.00e+00	0.00e+00	-1.71e-01	0.00e+00
Vth:	-9.01e-01	-9.07e-01	-9.01e-01	-9.07e-01	-9.07e-01	Vth:	-9.01e-01	-9.01e-01	-9.01e-01	7.11e-01	7.10e-01
Vdsat:	-2.99e-01	-2.99e-01	-2.99e-01	-2.97e-01	-2.97e-01	Vdsat:	-3.02e-01	-3.02e-01	-3.02e-01	1.70e-01	3.42e-01
Gm:	4.16e-05	4.16e-05	4.16e-05	4.19e-05	4.08e-12	Gm:	2.16e-04	2.09e-05	2.09e-05	7.33e-05	3.64e-05
Gds:	6.14e-08	6.14e-08	6.14e-08	6.05e-08	4.08e-05	Gds:	3.20e-07	3.12e-08	3.12e-08	6.07e-08	6.15e-08
Gmb:	2.68e-07	1.55e-07	1.56e-07	1.52e-14	1.52e-16	Gmb:	1.39e-06	1.35e-07	1.35e-07	4.17e-07	2.35e-07
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	Cgs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	1.52e-16	1.52e-16	1.52e-16	1.52e-16	1.52e-16	Cgsov:	3.21e-15	7.61e-17	7.61e-17	1.07e-16	2.66e-17
Cgdov:	1.52e-16	1.52e-16	1.52e-16	1.52e-16	1.52e-16	Cgdov:	3.21e-15	7.61e-17	7.61e-17	1.07e-16	2.66e-17
Cgbov:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	Cgbov:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgs:	7.93e-15	7.93e-15	7.93e-15	7.93e-15	5.95e-15	Cgs:	6.87e-13	3.97e-15	3.97e-15	5.55e-15	1.39e-15
Cgd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	5.95e-15	Cgd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgb:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	Cgb:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00

Name:	m:xl:7	m:xl:5	m:xl:2	m:xl:1
Model:	xl:n	xl:n	xl:n	xl:n
Id:	3.26e-05	6.32e-06	3.16e-06	3.16e-06
Vgs:	1.05e+00	1.05e+00	8.69e-01	8.69e-01
Vds:	3.21e+00	2.63e+00	1.17e+00	1.17e+00
Vbs:	0.00e+00	0.00e+00	-2.63e+00	-2.63e+00
Vth:	7.10e-01	7.10e-01	7.20e-01	7.20e-01
Vdsat:	3.42e-01	3.42e-01	1.48e-01	1.48e-01
Gm:	1.91e-04	3.70e-05	4.26e-05	4.26e-05
Gds:	3.16e-07	6.15e-08	3.12e-08	3.12e-08
Gmb:	1.23e-06	2.39e-07	1.18e-07	1.18e-07
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	1.37e-16	2.66e-17	7.17e-17	7.17e-17
Cgdov:	1.37e-16	2.66e-17	7.17e-17	7.17e-17
Cgbov:	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgs:	7.12e-15	1.39e-15	3.74e-15	3.74e-15
Cgd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgb:	0.00e+00	0.00e+00	0.00e+00	0.00e+00

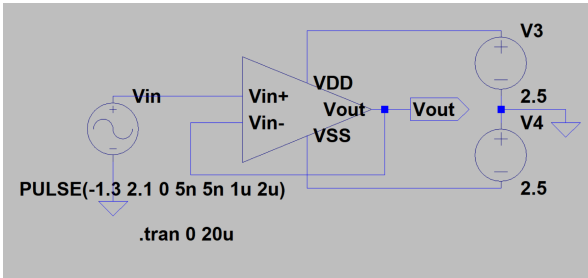
4-2- AC Analysis (.ac): The .ac simulation used a differential input: Vin+ with **1 V DC and 1 V AC**, and Vin- with **1 V DC only**. This setup allowed accurate open-loop small-signal analysis. Results showed a **DC gain of ~107 dB**, **phase margin ~62°**, and **GBW ≈ 5 MHz** after adjusting the target from 5 MHz to 5.5 MHz to account for nonidealities.



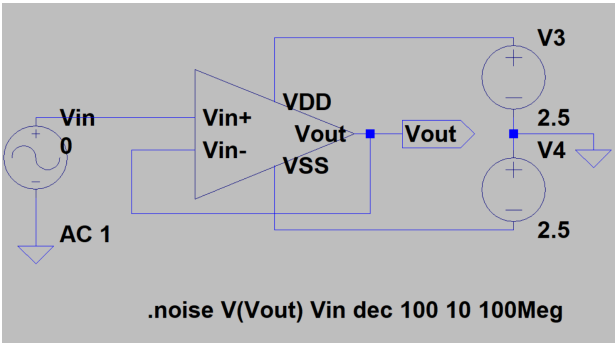
4-3- ICMR and Output Swing Analysis: A DC sweep on Vin+ from -1.3 V to 2.1 V confirmed that the input common-mode range meets the spec. In a closed-loop setup with R1 = 100 kΩ and R2 = 1 MΩ, a .dc sweep showed the output swings from -2.5 V to +2.5 V, exceeding the ±2.2 V requirement.



4-4- Slew Rate Analysis: A pulse input from -1.3 V to 2.1 V was applied to Vin+ in a .tran simulation. The output slew rates were approximately +5.8 V/μs (rising) and -5 V/μs (falling), satisfying the ±5 V/μs requirement.

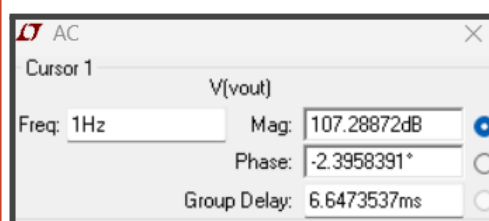
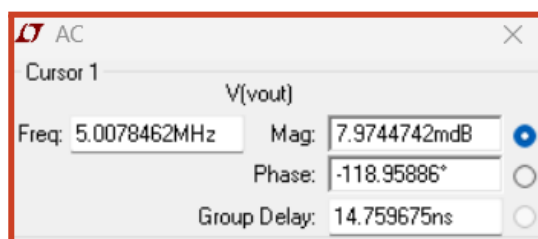
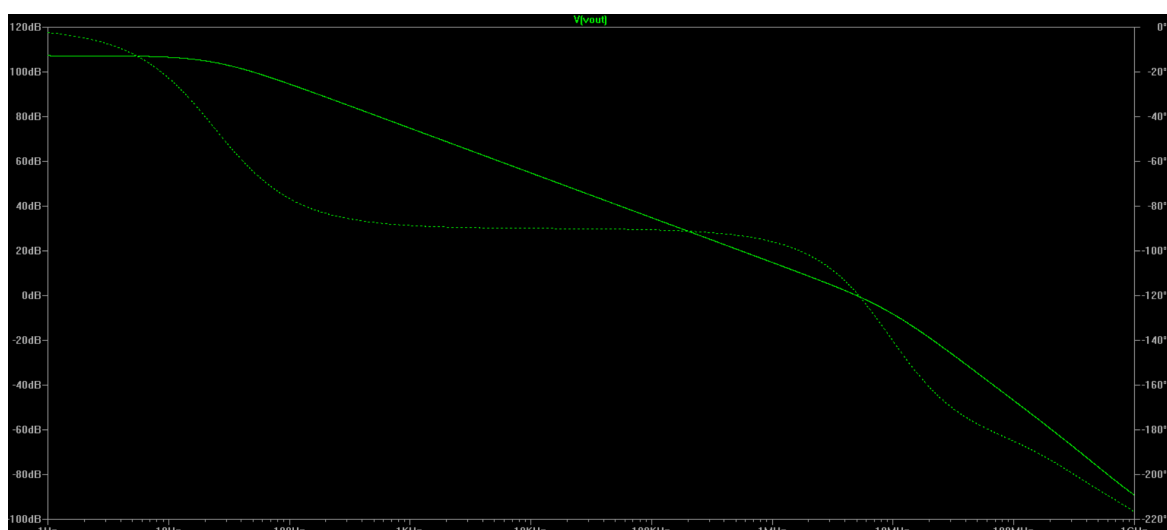


4-5- Noise Analysis: The .noise simulation was used to compute the output noise referred to the input. A 1 V AC signal was applied at Vin+, with Vin- grounded. The **input-referred noise** was found to **peak at 27.8 nV/√Hz**, successfully meeting the project specification of ≤ 30 nV/√Hz.

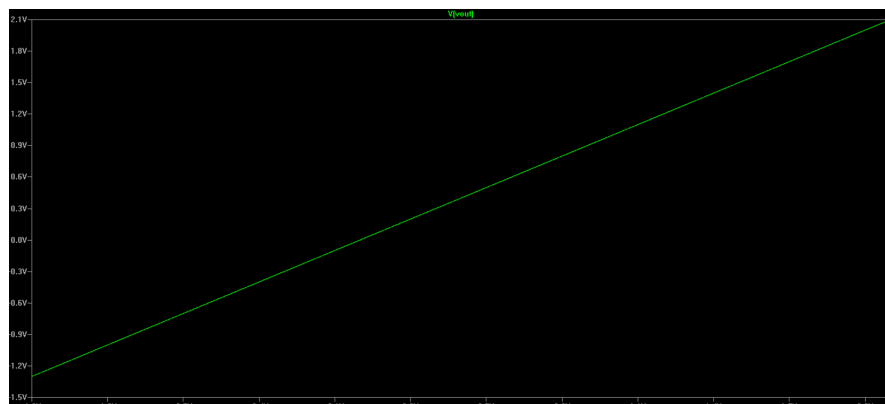


SPICE plots:

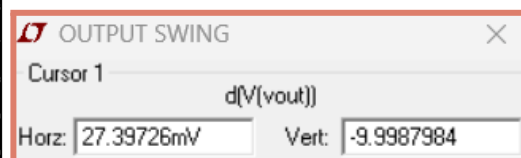
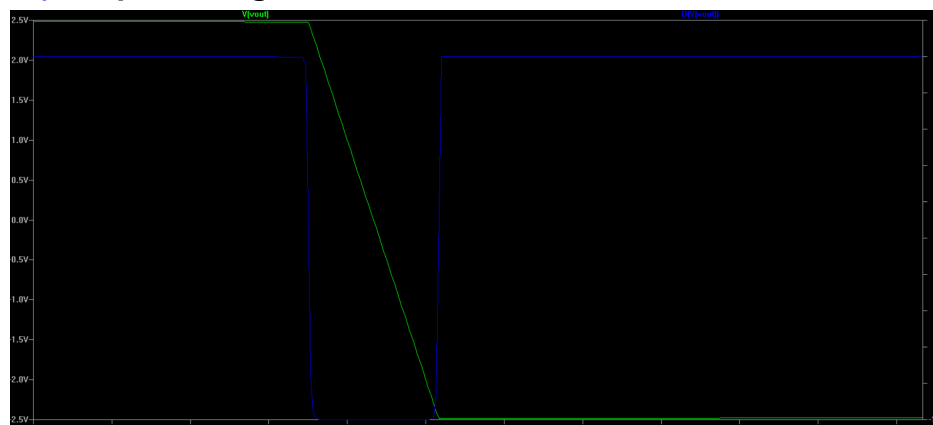
1) Bode Plot – (Gain vs Frequency from .ac) / Phase Plot – (Phase vs Frequency from .ac)



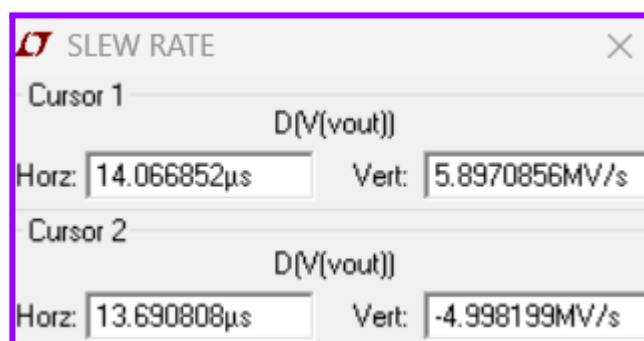
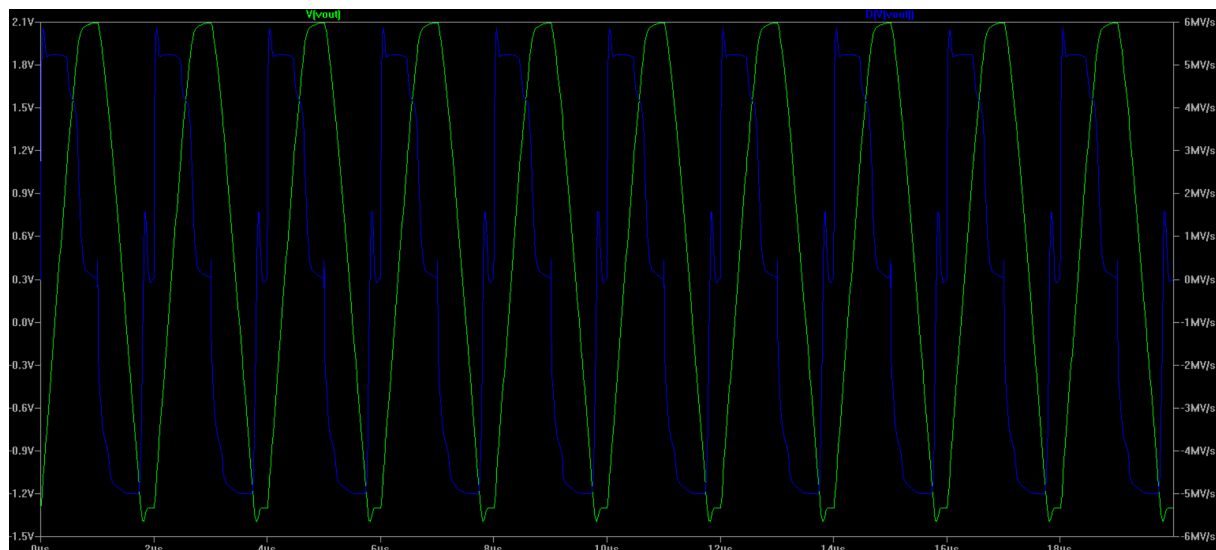
2) Input Common-Mode Range Plot ICMR Plot



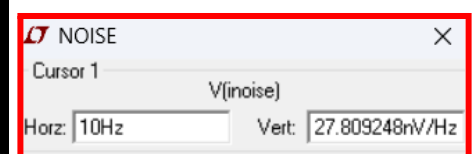
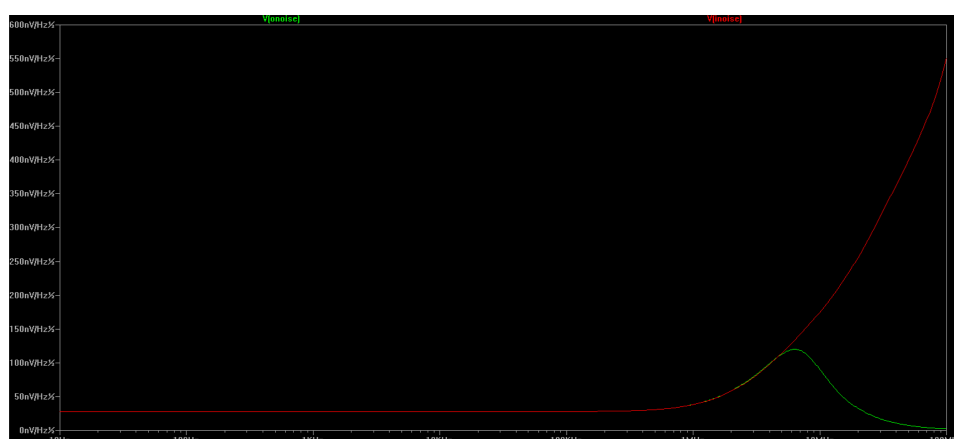
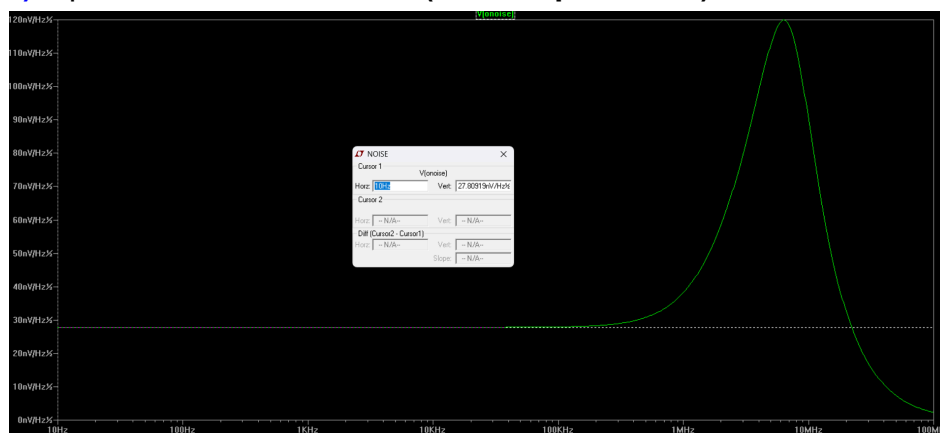
3) Output Swing Plot



4) Transient Slew Rate Plot



5) Input-Referred Noise Plot (and Output Noise)



Specification Comparison Table

Parameter	Specification	Achieved	Met?
DC Gain	≥ 80 dB	107 dB	✓
Phase Margin	$\geq 60^\circ$	62°	✓
Gain Bandwidth	≥ 5 MHz	5 MHz	✓
Slew Rate	$\geq \pm 5$ V/ μ s	+5.8 V/ μ s, -5 V/ μ s	✓
Input-referred Noise	≤ 30 nV/ $\sqrt{\text{Hz}}$	27.8 nV/ $\sqrt{\text{Hz}}$	✓
ICMR	-1.3 V to +2.1 V	-1.3 V to +2.1 V	✓
Output Swing	$\geq \pm 2.2$ V	± 2.5 V	✓